

linkit-v4

This variant integrates the SMD Argos module, optimized power management (wireless charging, solar input, and USB/Li-ion handling), and a compact architecture prioritizing energy efficiency, cost, and reliability. All interfaces and components are arranged to ensure seamless compatibility with the Linkit firmware ecosystem and Arribada's open-source tooling.

Sheet 1: Project overview (this page)

Sheet 2: linkit-v4

Sheet 3: pwr

Sheet 4: BQ51003YFPT_WirelessCharger_2.5W (devicesheet)

Sheet 5: TPS63901_BuckBoost_lowIq_3V32V3_400mA (devicesheet)

Sheet 6: mcu

Sheet 7: sensors

Sheet 8: BMA400_ACC_LinkitV3 (devicesheet)

Sheet 9: SAM-M10Q (devicesheet)

Sheet 10: interface

Sheet 11: satellite

Sheet 12: sat-smd

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

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This hardware design is an open-source project released under the CERN Open Hardware Licence Version 2 – Permissive (CERN-OHL-P v2).

You are free to:

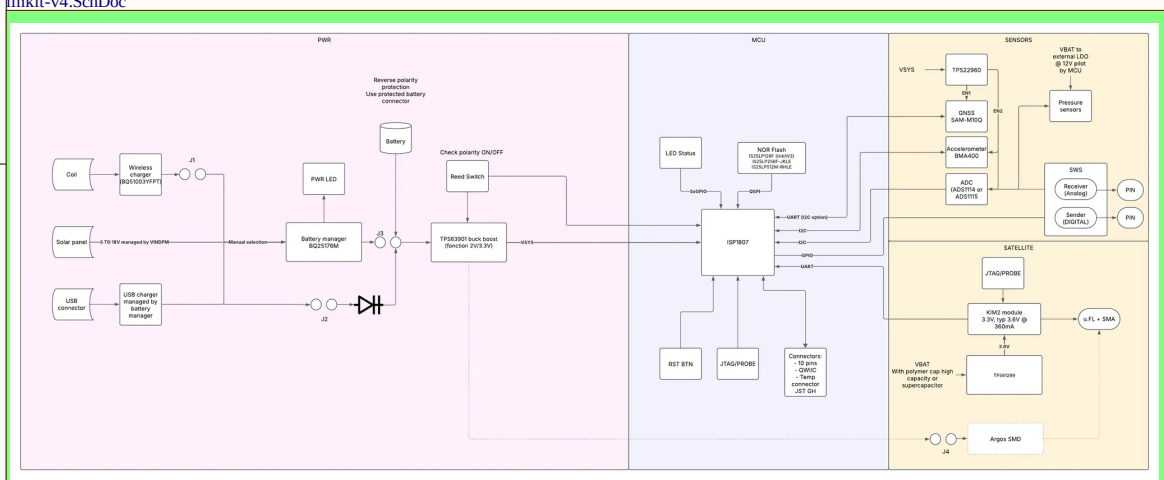
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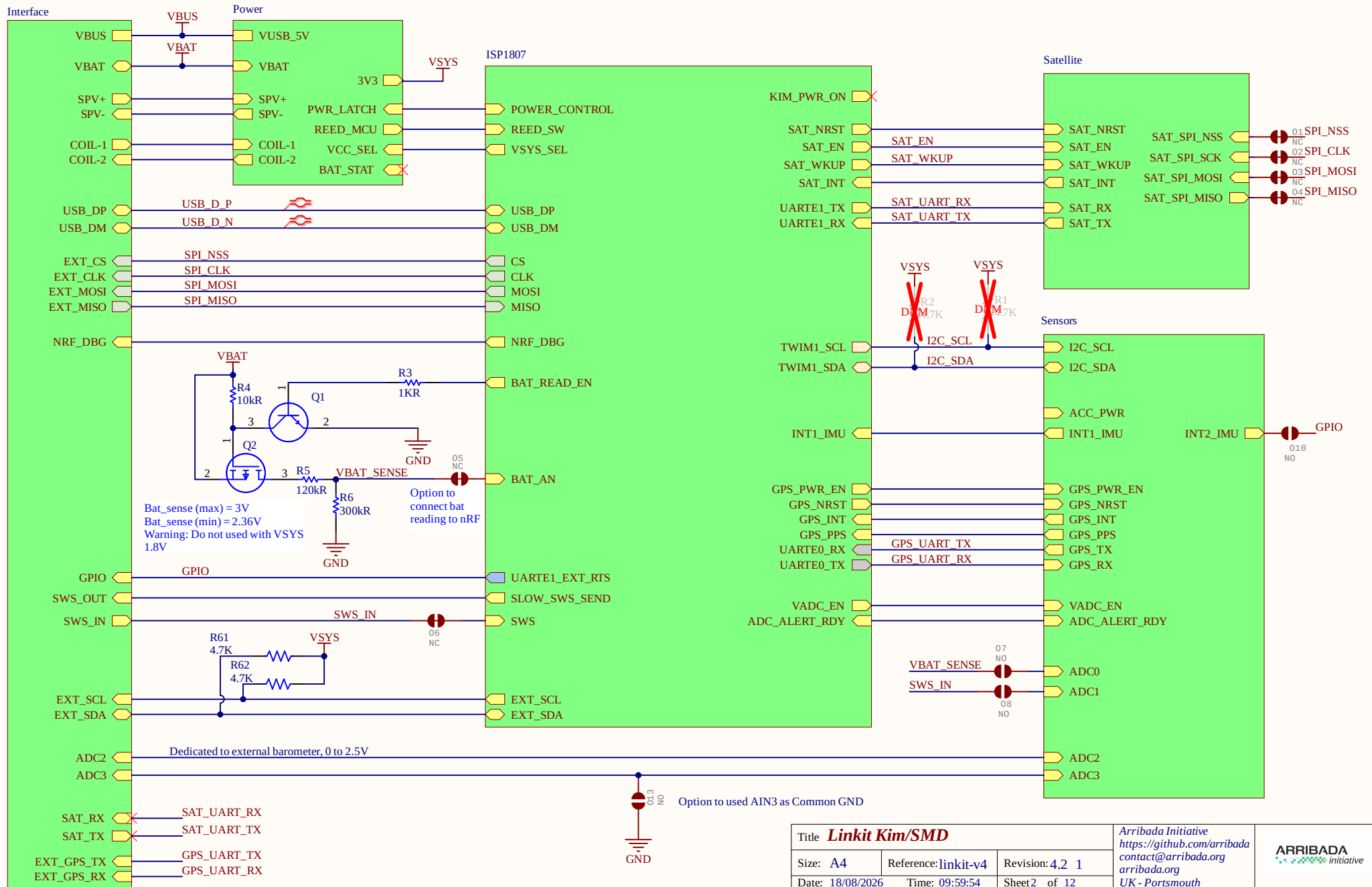
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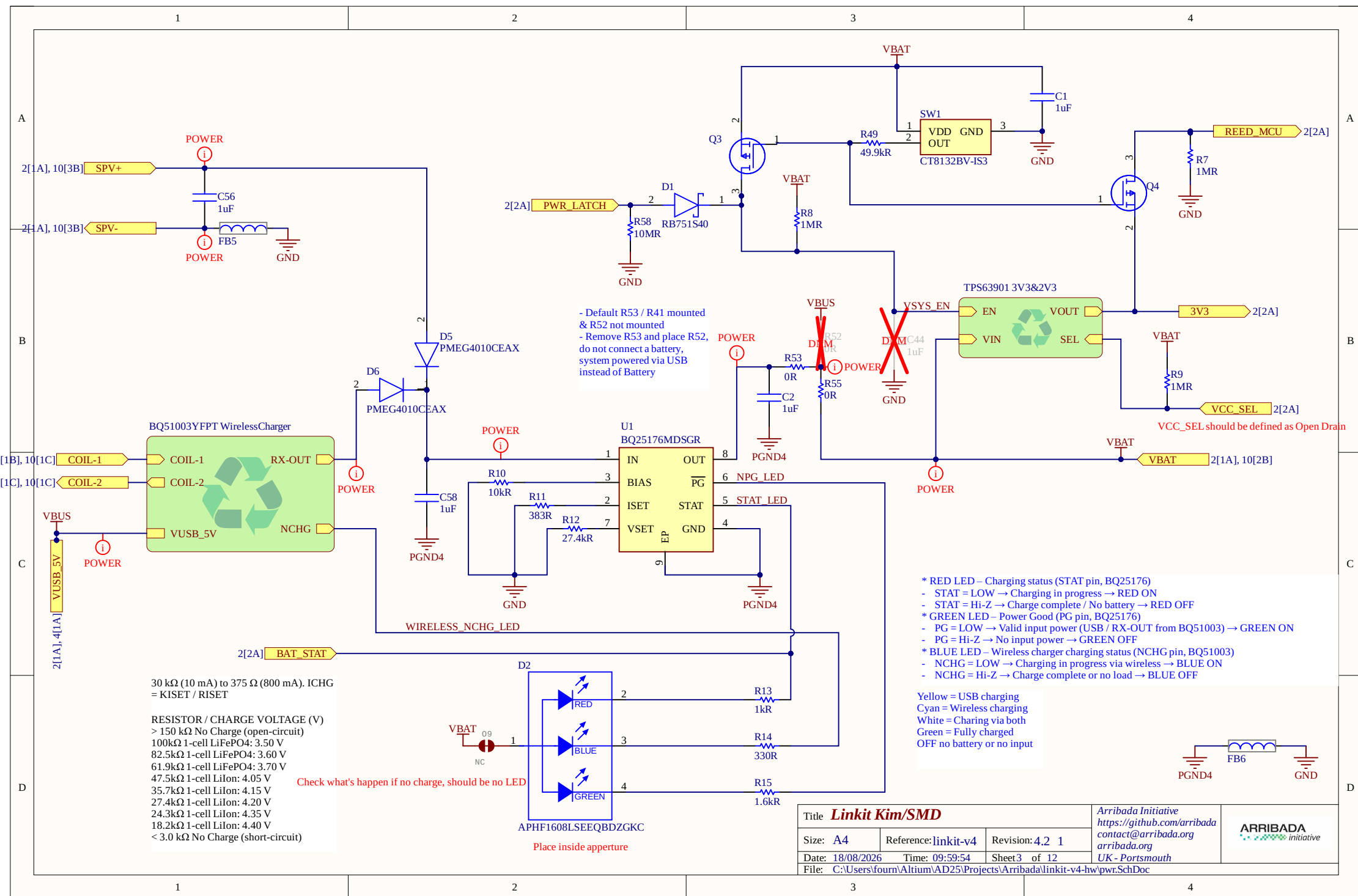
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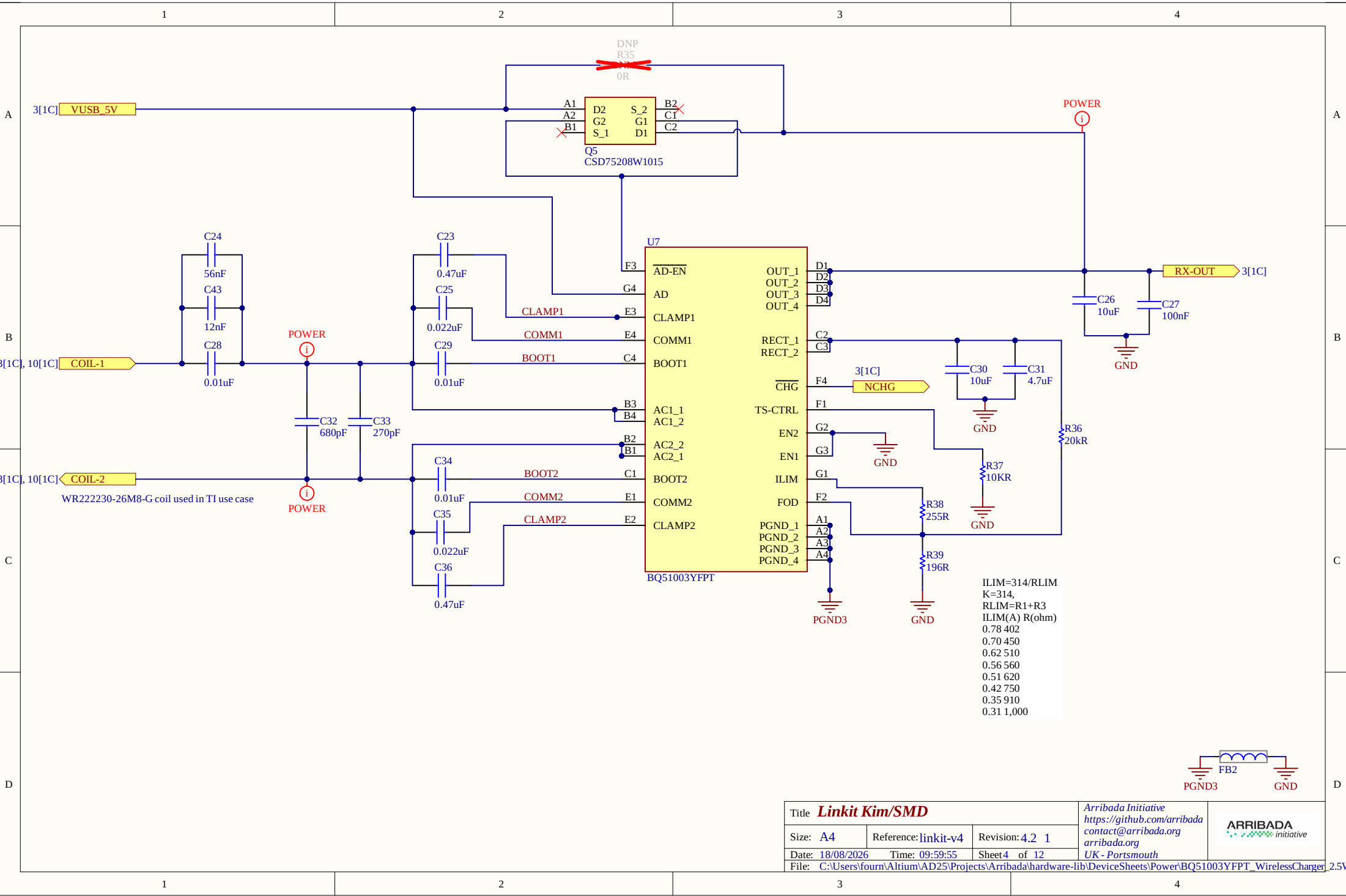


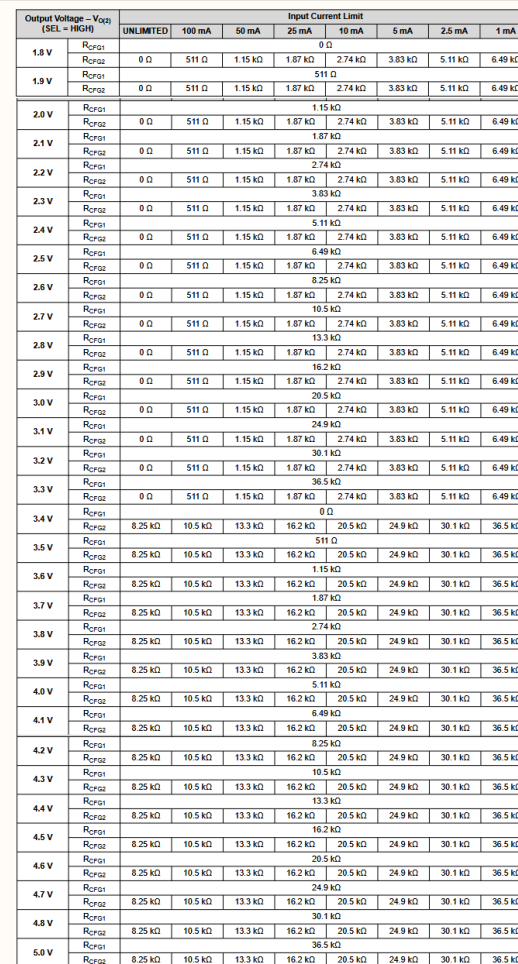
Title <i>Linkit Kim/SMD</i>			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth	
Size: A4	Reference: linkit-v4	Revision: 4.2 1		
Date: 18/08/2026	Time: 09:59:54	Sheet 1 of 12		
File: C:\Users\fourm\Altium\AD25\Projects\Arribada\linkit-v4-hw\Project Overview.SchDoc				



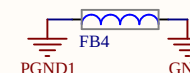
Title Linkit Kim/SMD			Arribada Initiative https://github.com/arribada contact@arribada.org arribada.org UK - Portsmouth 
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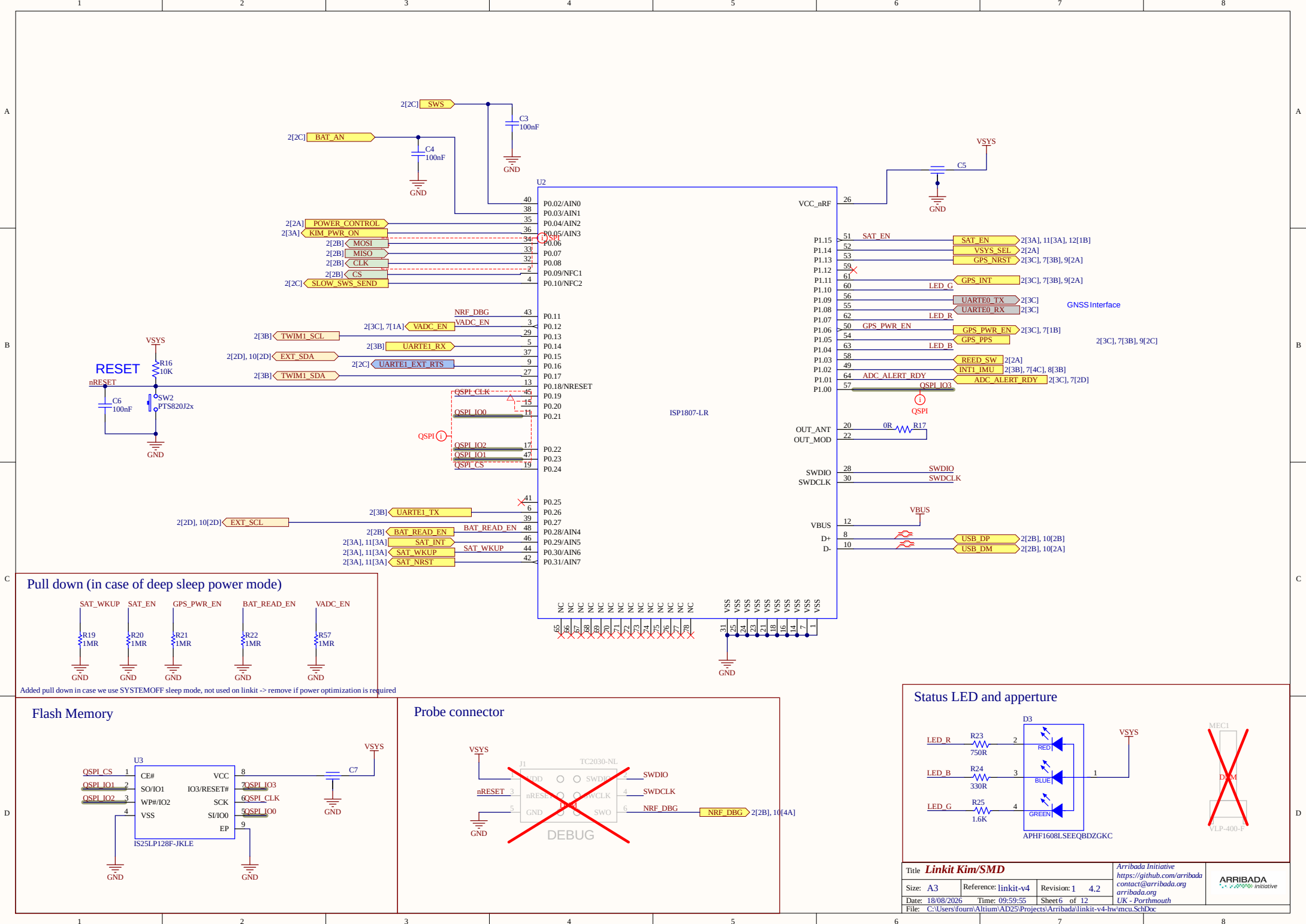




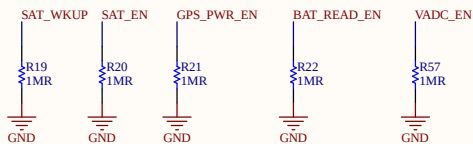


Output Voltage – V _{OUT} (SEL = LOW)	R _{PROCESS}
1.8 V	0 Ω
2.0 V	511 Ω
2.1 V	1.15 kΩ
2.2 V	1.87 kΩ
2.3 V	2.74 kΩ
2.4 V	3.83 kΩ
2.5 V	5.11 kΩ
2.6 V	6.49 kΩ
2.7 V	8.25 kΩ
2.8 V	10.5 kΩ
3.0 V	13.3 kΩ
3.3 V	16.2 kΩ
3.6 V	20.5 kΩ
4.0 V	24.9 kΩ
4.5 V	30.1 kΩ
5.0 V	36.5 kΩ



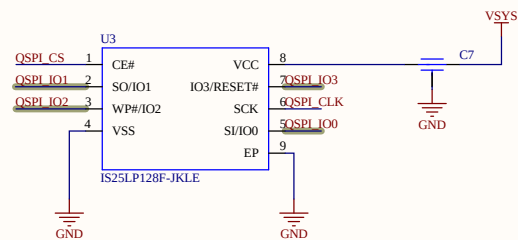


Pull down (in case of deep sleep power mode)

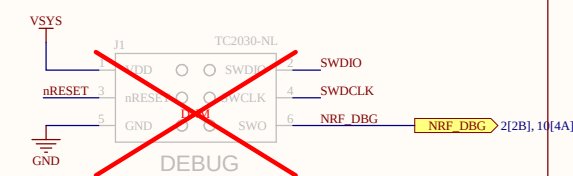


Added pull down in case we use SYSTEMOFF sleep mode, not used on linkit -> remove if power optimization is required

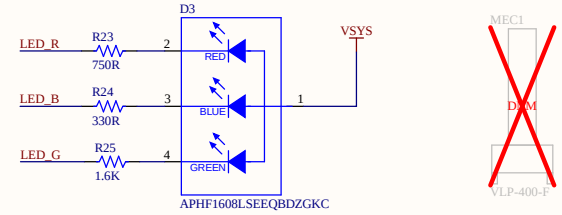
Flash Memory

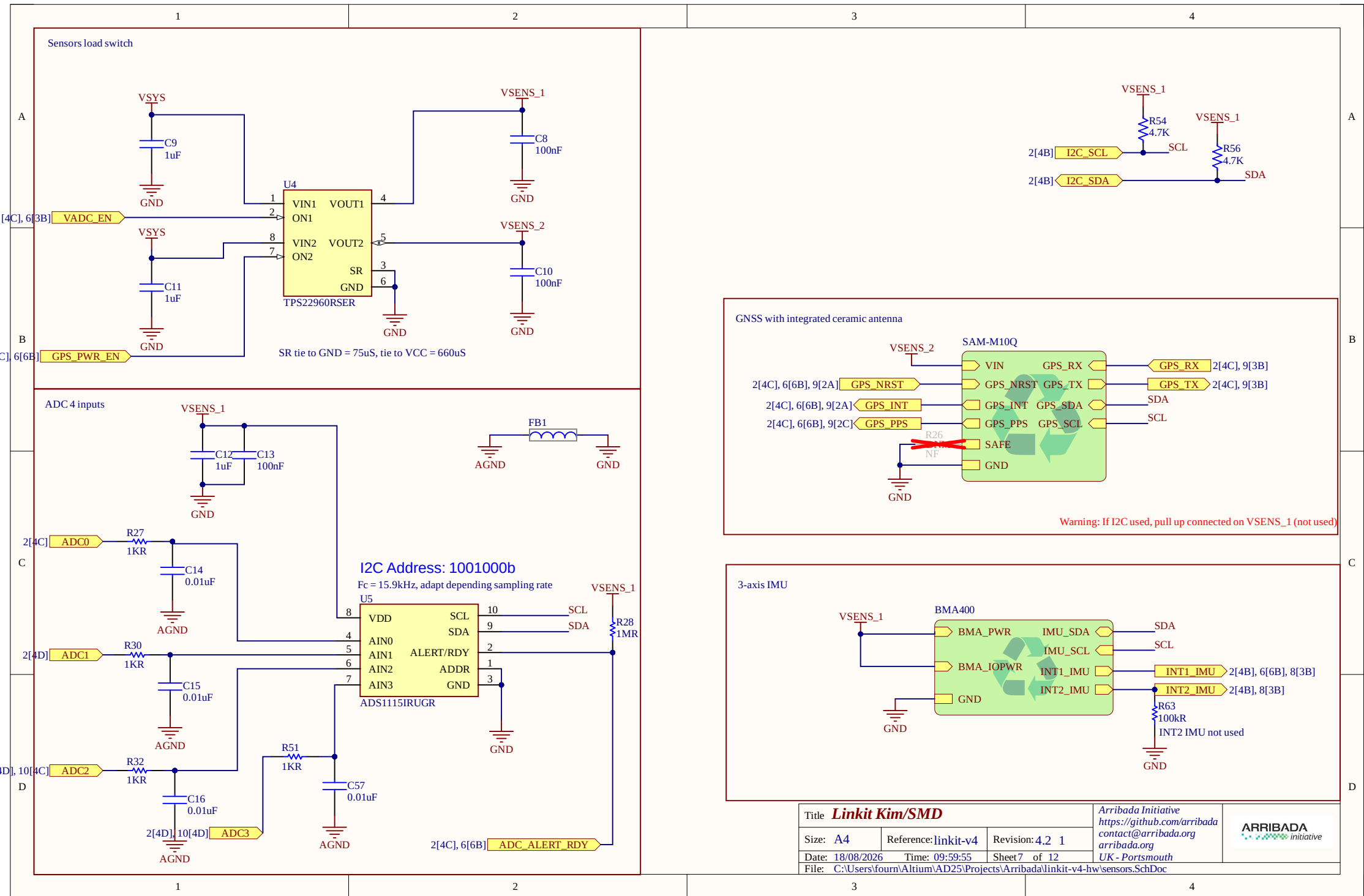


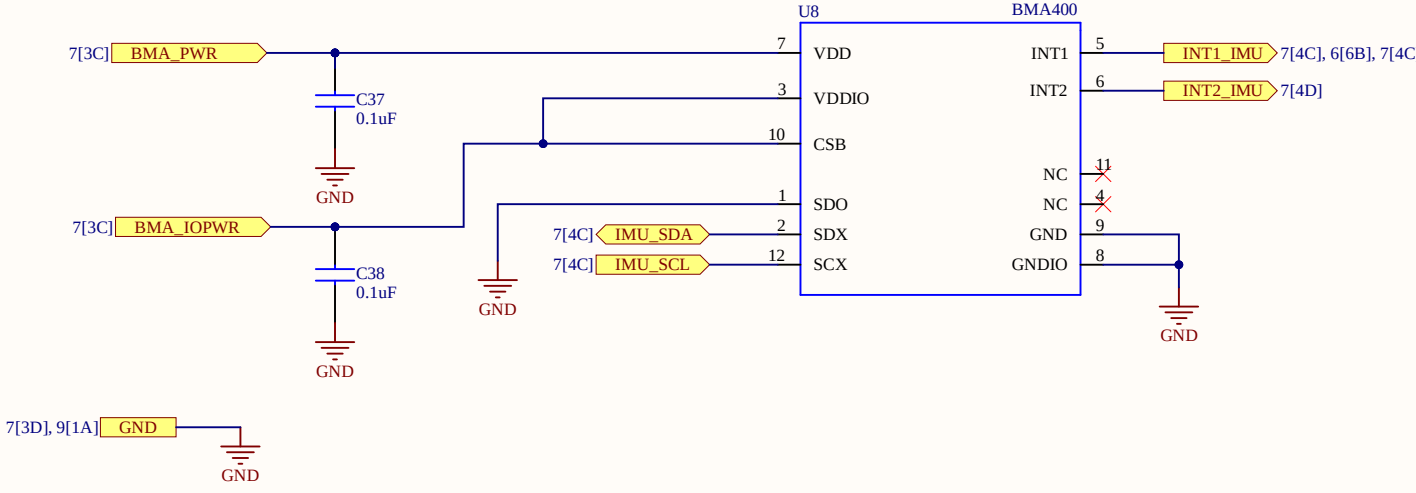
Probe connector



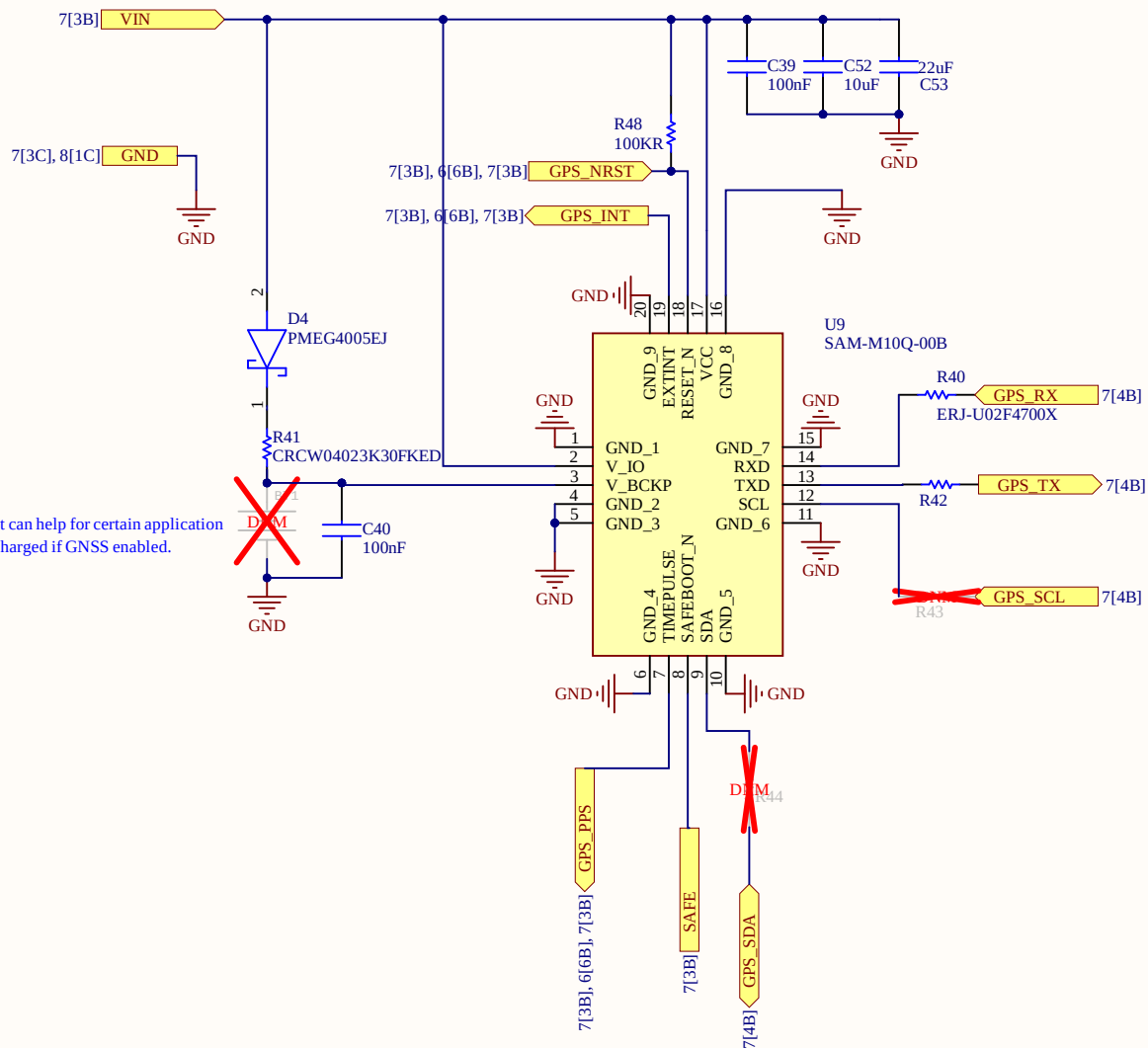
Status LED and aperture





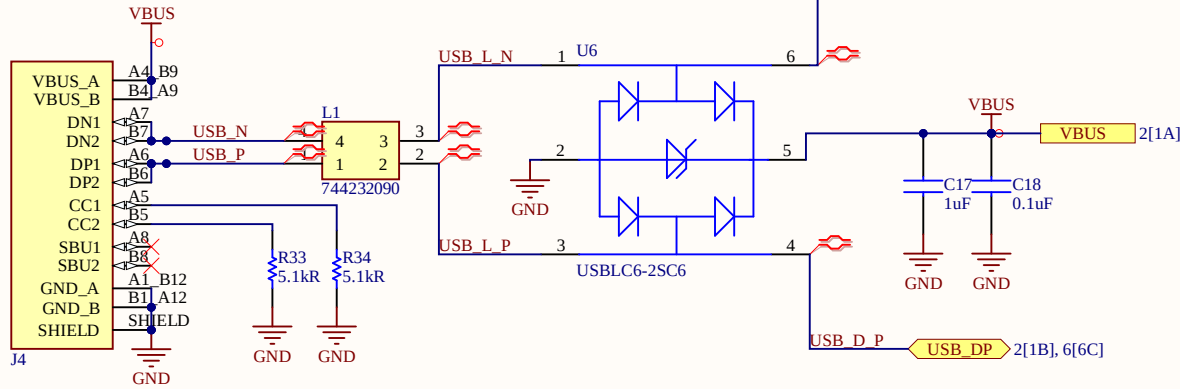


Option, not required but can help for certain application
Cell backup coin only charged if GNSS enabled.

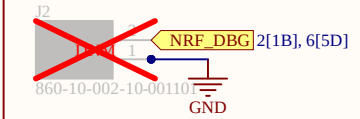


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File: C:\Users\fourm\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\GPS\SAM-M10Q.SchDoc					

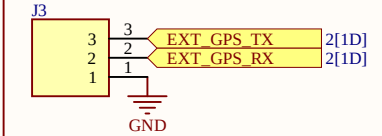
USB Type-C



NRF Debug UART out

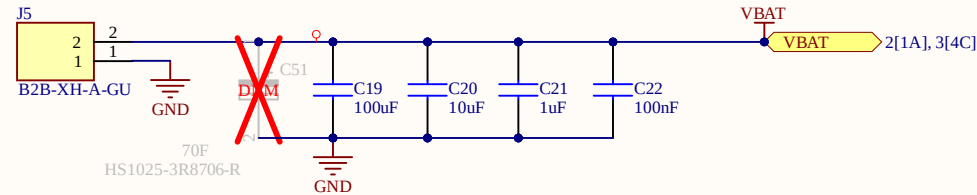


GPS Debug



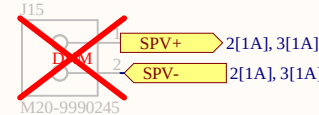
Battery and decoupling

Add polarity indicator

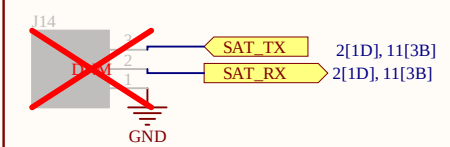


Solar panel input

Add polarity indicator

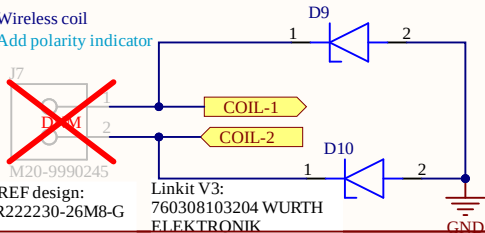


SAT UART



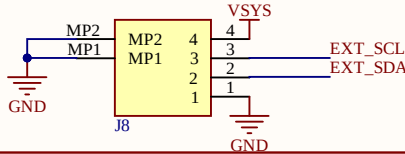
Wireless coil

Add polarity indicator



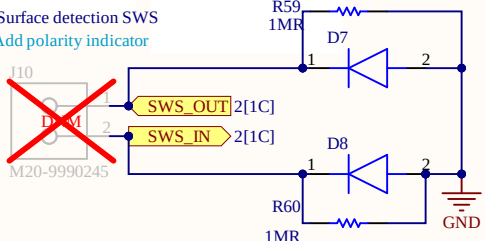
BM04B connector dedicated to bluerobotics thermal sensor

Add pin definition



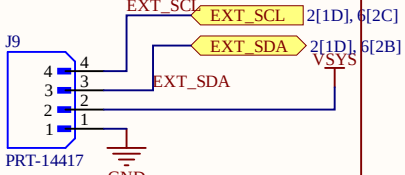
Surface detection SWS

Add polarity indicator

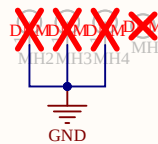


Qwiic I2C connector

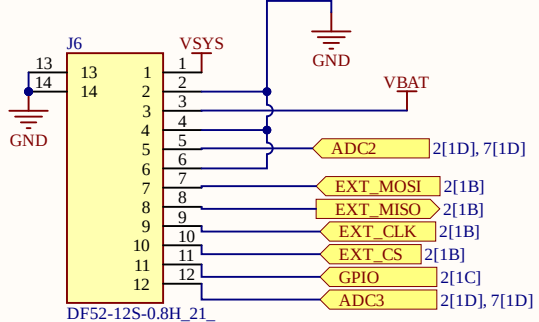
Add pin definition



Fixation VIAs



Header interface



Title **Linkit Kim/SMD**

Size: **A4**

Reference: **linkit-v4**

Revision: **4.2 1**

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Sheet **10** of **12**

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 UK - Portsmouth



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1

2

3

4

A

A

B

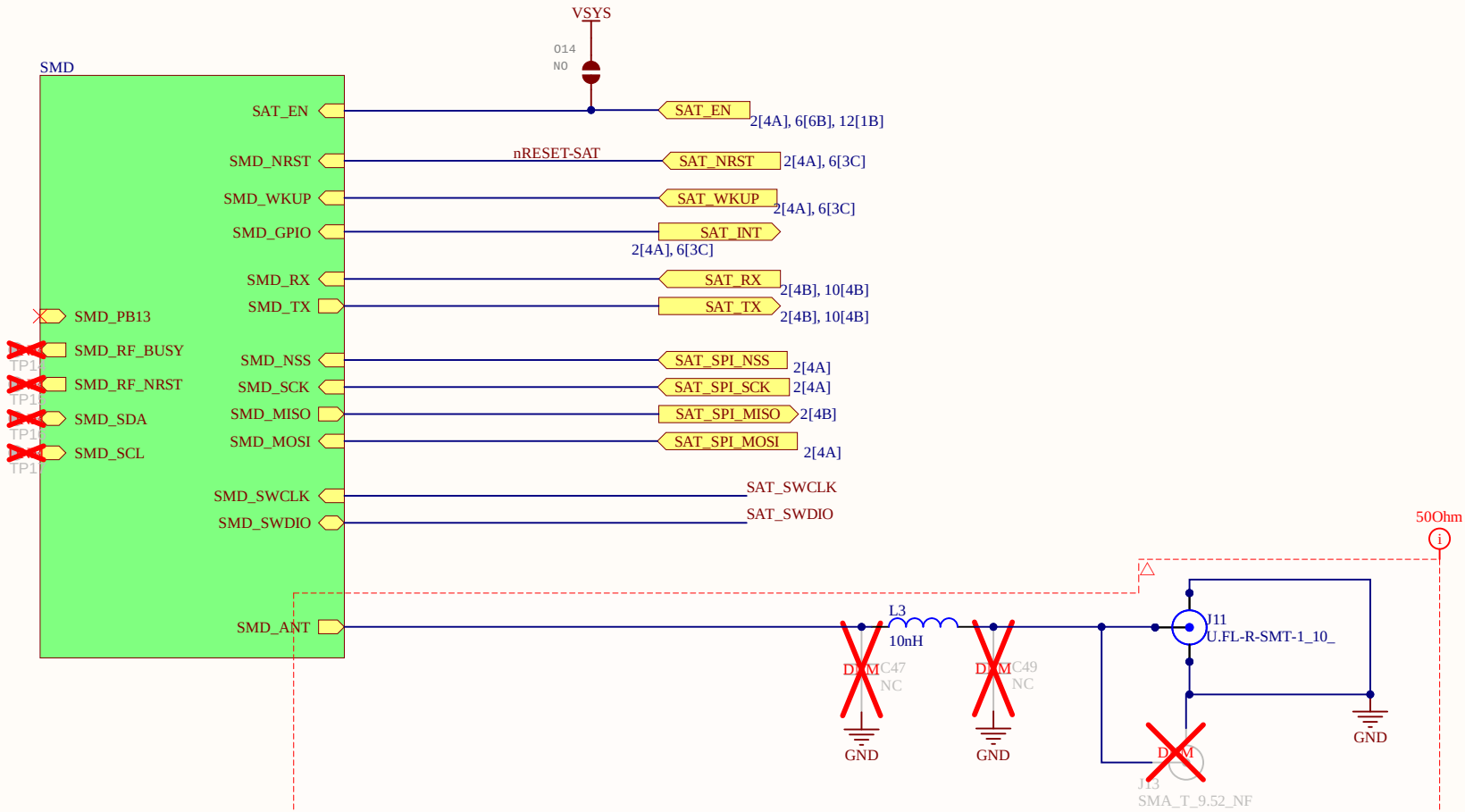
B

C

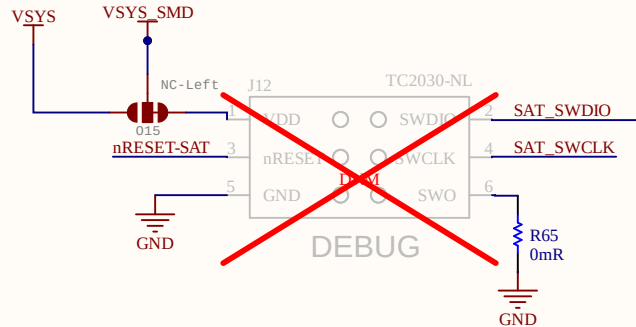
C

D

D



Satellite Debugger

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Revision: 4.2 1

Date: 18/08/2026

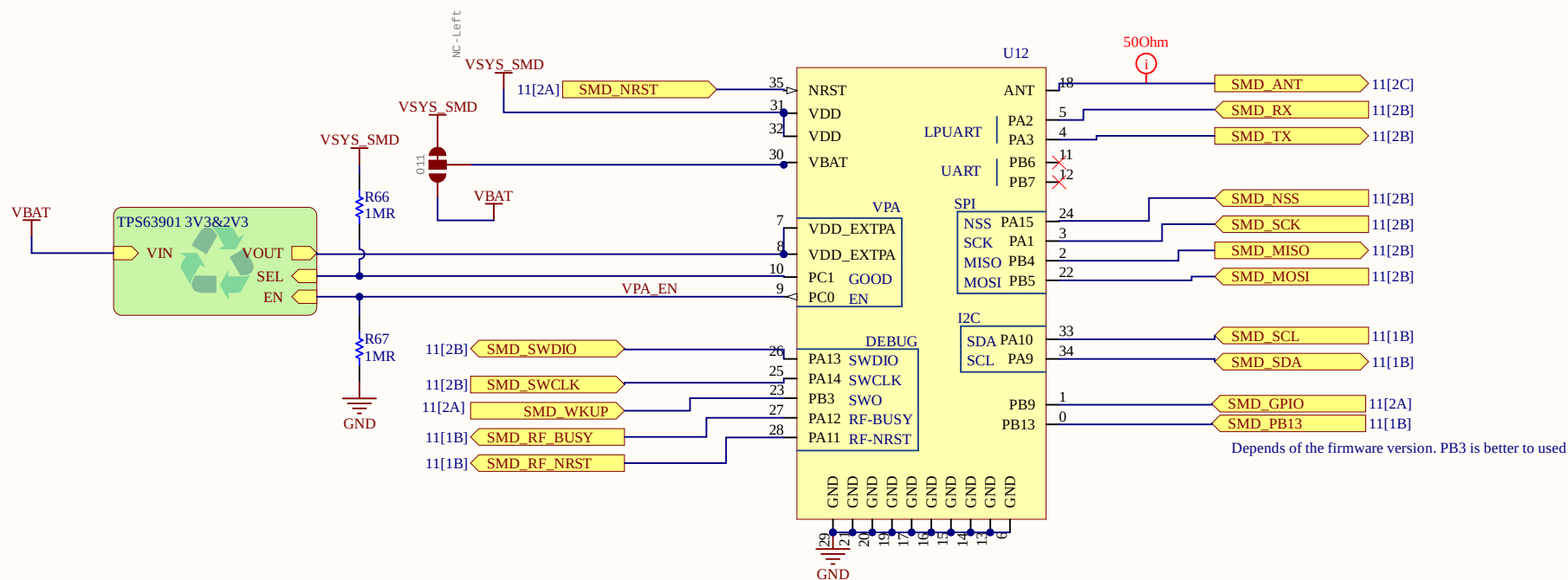
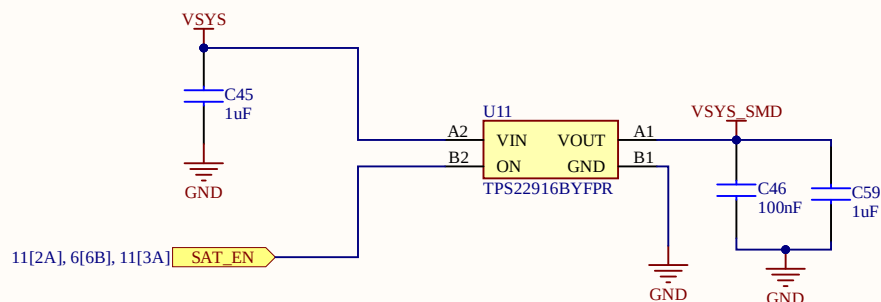
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